

INNOVA JUNIOR COLLEGE
 JC 2 PRELIMINARY EXAMINATION 2
 in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY

9647/01

Paper 1 Multiple Choice

18 September 2013

1 hour

Additional Materials: Data Booklet
 Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
 Write in soft pencil.
 Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.
 Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
 Any rough working should be done in this booklet.

This document consists of **18** printed pages.



Section A

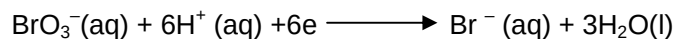
For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1 A giant molecule contains a large amount of carbon, mainly of isotopes ^{12}C and ^{13}C . It was found that the relative atomic mass of carbon in the molecule is 12.20.

What is the ratio of ^{12}C and ^{13}C ?

- A** 3:1 **B** 3:4 **C** 4:1 **D** 1:4

- 2 20.0 cm^3 of 0.02 mol dm^{-3} aqueous sodium bromate(V), NaBrO_3 , was found to react completely with 80.0 cm^3 of 0.01 mol dm^{-3} hydroxylamine, NH_2OH . The half equation for the reduction of bromate(V) ions is given as shown.



Which of the following could be the nitrogen containing product in this reaction?

- A** N_2O
B NO_2
C NO
D NO_3^-

- 3 The successive ionisation energies (IE) of two elements, **A** and **B**, are given below.

IE/ kJ mol^{-1}	1st	2nd	3rd	4th	5th	6th	7th	8th
A	1090	2350	4610	6220	37800	47000	-	-
B	1251	2298	3822	5158	6542	9362	11018	33604

What is the likely formula of the compound that is formed when **A** reacts with **B**?

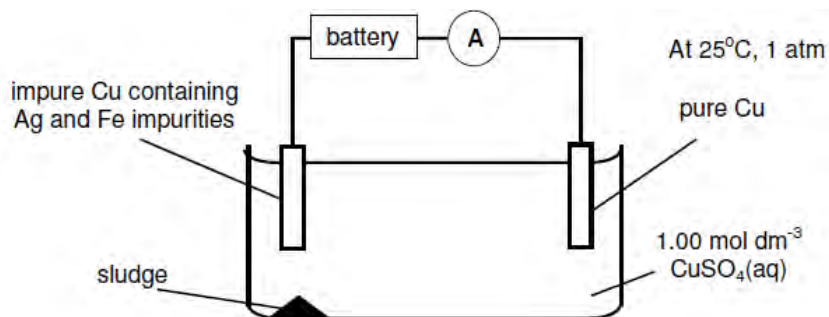
- A** **AB** **B** **A_2B_3** **C** **AB_4** **D** **A_4B**

- 4 Which electronic configuration represents an atom of an element that forms a simple ion with a charge of -3 ?

- A** $1s^2 2s^2 2p^6 3s^2 3p^1$
B $1s^2 2s^2 2p^6 3s^2 3p^3$
C $1s^2 2s^2 2p^6 3s^2 3p^6 3d^1 4s^2$
D $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$

- 5 Which of the following species has a square planar structure?
- A BrF_4^-
 B BF_4^-
 C CH_4
 D C_2H_4
- 6 Which of the following processes is endothermic?
- A condensation of steam
 B freezing of water
 C electrolysis of water
 D $\text{Ca(s)} + 2\text{H}_2\text{O(l)} \longrightarrow \text{Ca(OH)}_2\text{(aq)} + \text{H}_2\text{(g)}$
- 7 Which value would be required to estimate the lattice energy for the hypothetical ionic compound MgH_2 ?
- A The electron affinity of hydrogen
 B The first ionisation energy of hydrogen
 C The magnesium-hydrogen bond energy
 D The standard enthalpy change of formation of MgH
- 8 *Use of the Data Booklet is relevant to this question.*

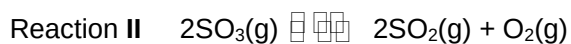
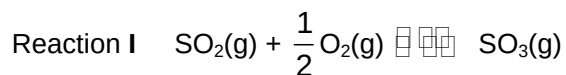
The circuit shown in the diagram was set up.



What are the products formed at the anode and cathode and what is the substance present in the sludge after some time?

	anode	cathode	sludge
A	Cu^{2+}	Cu	Ag
B	Fe^{2+}	Cu^{2+}	Ag
C	$\text{Ag}^+, \text{Fe}^{2+}$	Cu^{2+}	Zn
D	$\text{Cu}^{2+}, \text{Fe}^{2+}$	Cu	Ag

- 9 The equilibrium constant for reaction I below is K . What is the equilibrium constant for reaction II?



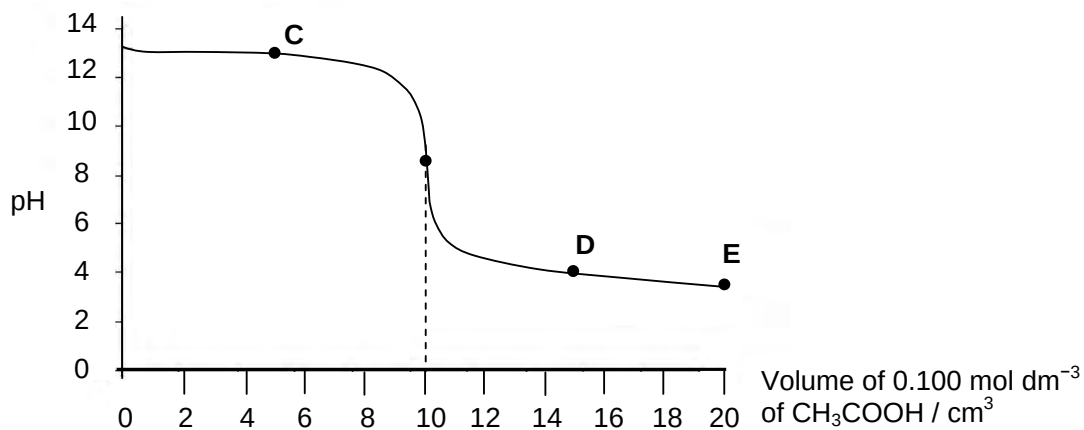
A K^{-2}

C $2K$

B K^2

D $2K^{-1}$

- 10 When 5.00 cm^3 of a $0.100 \text{ mol dm}^{-3}$ base is titrated against $0.100 \text{ mol dm}^{-3}$ of aqueous CH_3COOH , the following titration curve is obtained.



Which of the following correctly identifies the base used and the point on the curve at which $\text{pH} = \text{pK}_a$?

	base used	point where $\text{pH} = \text{pK}_a$
A	$\text{KOH}(\text{aq})$	C
B	$\text{Ba}(\text{OH})_2(\text{aq})$	D
C	$\text{KOH}(\text{aq})$	E
D	$\text{Ba}(\text{OH})_2(\text{aq})$	E

- 11 The table gives the concentrations and pH values of the aqueous solutions of two compounds, **C** and **D**. Either compound could be an acid or a base.

	C	D
Concentration	2 mol dm ⁻³	2 mol dm ⁻³
pH	6	9

Student **E** concluded that **C** is a strong acid.

Student **F** concluded that the extent of dissociation is lower in **C(aq)** than in **D(aq)**.

Which of the students is/are correct?

- A **E** only
 B **F** only
 C Both **E** and **F**
 D Neither **E** nor **F**
- 12 A 10 cm³ sample of 0.010 mol dm⁻³ HCl is diluted by adding distilled water at constant temperature.

Which one of the following items correctly shows the effect of the dilution on the concentrations of H⁺ and OH⁻ ions in the solution?

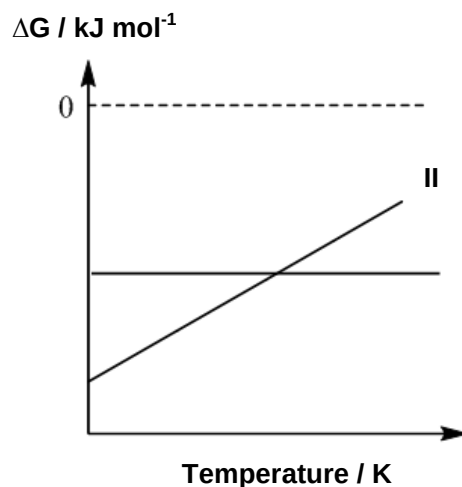
	[H ⁺]	[OH ⁻]
A	increase	increase
B	increase	decrease
C	decrease	increase
D	decrease	decrease

- 13 Lead is the final product formed by a series of changes in which the rate determining stage is the radioactive decay of uranium-238. This radioactive decay is a first order reaction with a half-life of 4.5×10^9 years.

What would be the age of a rock sample, originally lead-free, in which the molar proportion of uranium to lead is now 1:3?

- A 1.5×10^9 years
 B 2.25×10^9 years
 C 4.5×10^9 years
 D 9.0×10^9 years

- 14 Ellingham diagrams are plots of ΔG against temperature, T . An Ellingham diagram for two reactions involving the oxidation of S and SO_2 is as shown below:



Given that $\Delta G = \Delta H - T\Delta S$, which of the following correctly shows the two reactions corresponding to reactions I and II in the above Ellingham diagram?

	reaction I	reaction II
A	$2\text{S(s)} + 3\text{O}_2\text{(g)} \longrightarrow 2\text{SO}_3\text{(g)}$	$2\text{SO}_2\text{(g)} + \text{O}_2\text{(g)} \longrightarrow 2\text{SO}_3\text{(g)}$
B	$2\text{SO}_2\text{(s)} + \text{O}_2\text{(g)} \longrightarrow 2\text{SO}_3\text{(g)}$	$\text{S(s)} + \text{O}_2\text{(g)} \longrightarrow \text{SO}_2\text{(g)}$
C	$\text{S(s)} + \text{O}_2\text{(g)} \longrightarrow \text{SO}_2\text{(g)}$	$2\text{S(s)} + 3\text{O}_2\text{(g)} \longrightarrow 2\text{SO}_3\text{(g)}$
D	$2\text{SO}_2\text{(g)} + \text{O}_2\text{(g)} \longrightarrow 2\text{SO}_3\text{(g)}$	$2\text{S(s)} + 3\text{O}_2\text{(g)} \longrightarrow 2\text{SO}_3\text{(g)}$

- 15 When one mole sample of hydrated chromium(III) chloride, $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ is treated with excess of aqueous silver nitrate, 3 moles of silver chloride is precipitated. What is the most probable formula of the complex ion in the salt?

- A $[\text{CrCl}_2(\text{H}_2\text{O})_4]^+$
 B $[\text{CrCl}(\text{H}_2\text{O})_5]^{2+}$
 C $[\text{CrCl}(\text{H}_2\text{O})_3]^{2+}$
 D $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$

- Which of the following is correct in order of increasing proton number?

D **N, M, K, L**

- D** The number of electrons are the same but the number of protons increases.

- D** the tendency to form complex ions

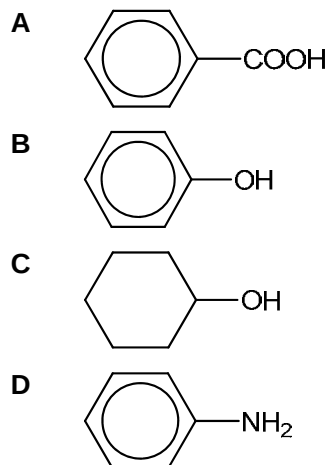
- D 8

- 20 Species with the molecular formula CH_3 can act as an electrophile, a free radical or a nucleophile depending on the number of outer shell electrons on the central carbon atom.

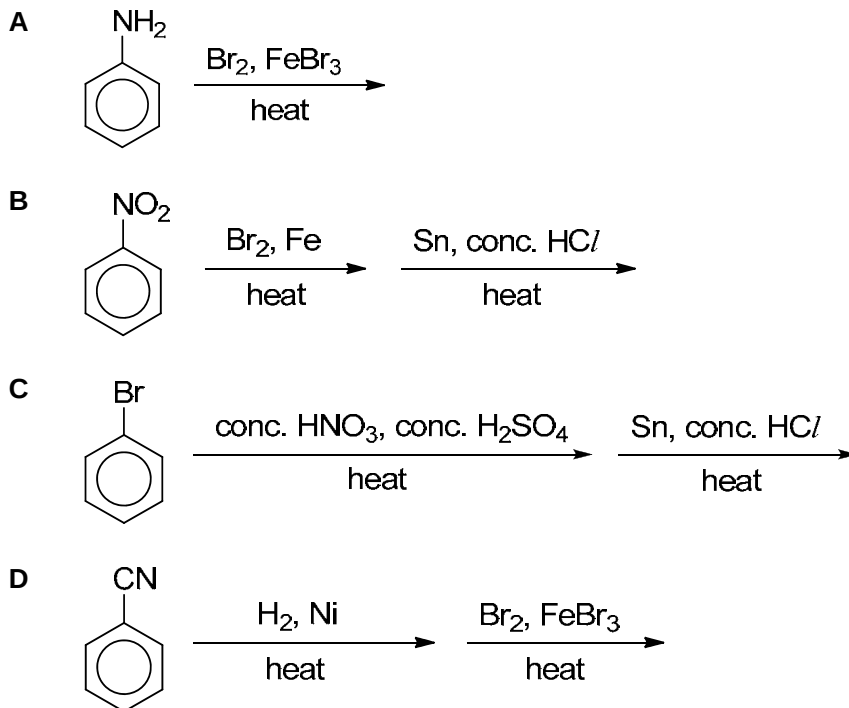
How many outer shell electrons must be present for CH_3 to act in these different ways?

	CH_3 as an electrophile	CH_3 as a free radical	CH_3 as a nucleophile
A	6	7	8
B	6	8	7
C	7	6	8
D	8	7	6

- 21 Compound **O** dissolves in aqueous sodium hydroxide to form a homogenous solution. When benzoyl chloride is added to the resulting solution, a white suspension forms. Which of the following could be compound **O**?



- 22 Which of the following reaction schemes produces 3-bromophenylamine in the highest yield?

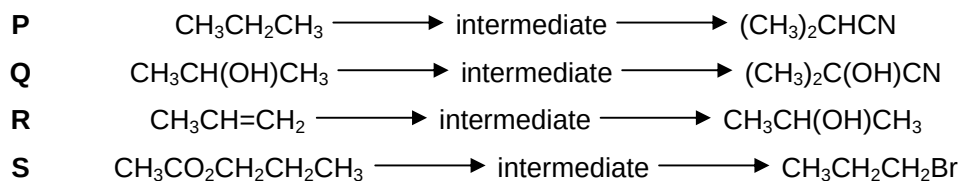


- 23 High energy radiation in the stratosphere produces free-radicals from chlorofluoroalkanes, commonly known as CFCs.

Which free radical is most likely to result from the irradiation of $\text{CHFClCF}_2\text{Cl}$?

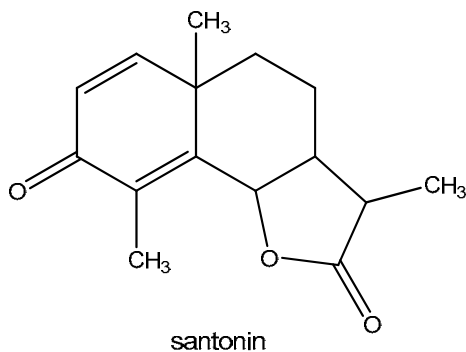
- A $\text{CHFCl}\dot{\text{C}}\text{FCl}$
- B $\dot{\text{C}}\text{HClCF}_2\text{Cl}$
- C $\dot{\text{C}}\text{HFCF}_2\text{Cl}$
- D $\dot{\text{C}}\text{FCF}_2\text{Cl}$

- 24 Which pair of reactions could have the same common intermediate?



- A P and Q
- B P and R
- C Q and S
- D R and S

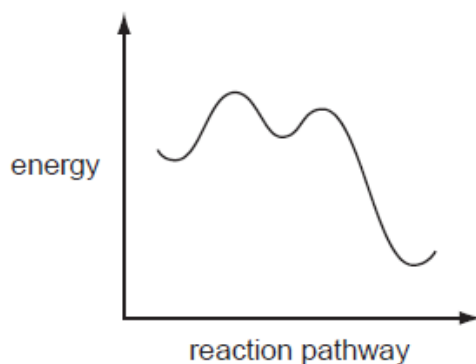
- 25 Santonin is a drug that was once widely used to expel parasitic worms from the body.



When santonin is first treated with warm dilute H_2SO_4 , and then the product of this reaction is treated with cold acidified KMnO_4 , a final product **T** is obtained.

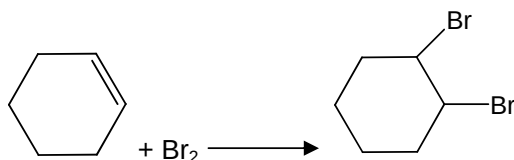
How many atoms of hydrogen in each molecule of product **T** can be displaced with sodium metal?

- | | | | |
|----------|---|----------|---|
| A | 2 | C | 5 |
| B | 4 | D | 6 |
- 26 A reaction pathway diagram is shown.

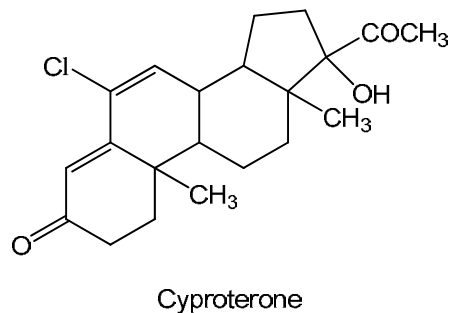
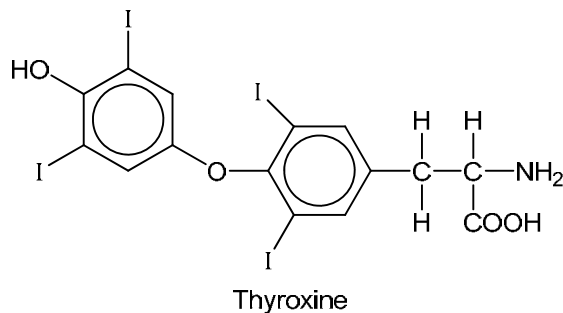


Which reaction does **not** have such a profile?

- A** $\text{CH}_3\text{CHO} + \text{HCN} \xrightarrow{\text{NaCN}} \text{CH}_3\text{CH}(\text{OH})\text{CN}$
- B** $\text{C}_2\text{H}_5\text{Br} + \text{NaOH} \longrightarrow \text{C}_2\text{H}_5\text{OH} + \text{NaBr}$
- C** $(\text{CH}_3)_3\text{CBr} + \text{NaOH} \longrightarrow (\text{CH}_3)_3\text{COH} + \text{NaBr}$
- D**



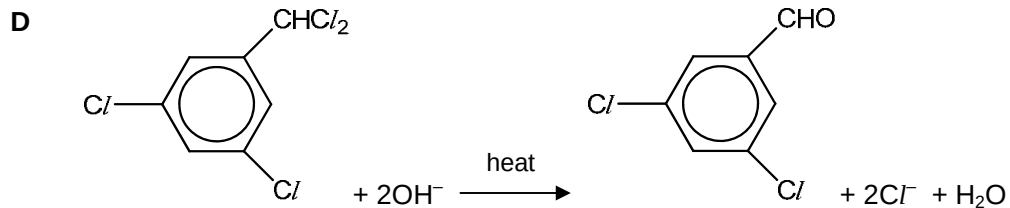
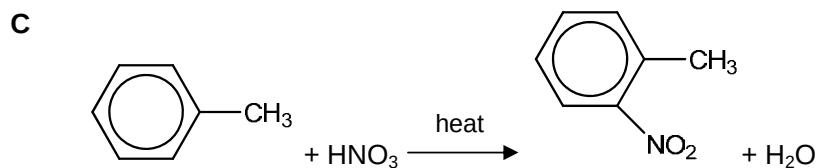
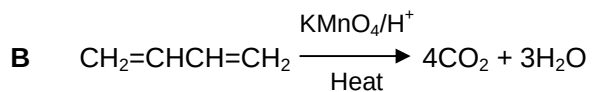
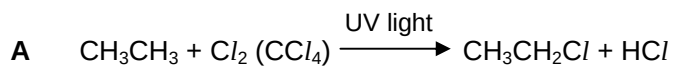
- 27 Thyroxine and Cyproterone are compounds that are often used in hormone therapy.



Which of the following reagents **cannot** be used to distinguish between the two compounds?

- | | |
|--|---|
| <p>A neutral FeCl_3</p> <p>B alkaline aqueous iodine</p> | <p>C aqueous silver nitrate</p> <p>D sodium carbonate</p> |
|--|---|
- 28 The reaction conditions for four different transformations are given below.

Which transformation has a set of conditions that is **not** correct?

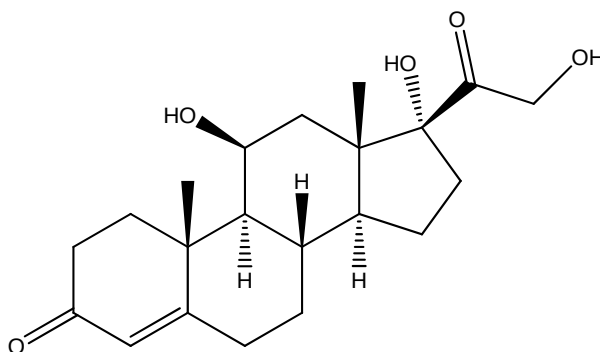


- 29 The C–H bond lengths of four hydrocarbons are given in the table below.

compound	structural formula	C–H bond length /nm
methane	CH ₄	0.110
ethane	CH ₃ CH ₃	0.110
ethene	CH ₂ =CH ₂	0.108
ethyne	CH≡CH	0.106

Which of the following helps to explain the shortest C–H bond length observed in ethyne?

- A Ethyne is a linear molecule.
- B The orbital of the carbon atom used in formation of the C–H bond in ethyne has the greatest s-orbital character.
- C An sp²-sp² overlap is observed between the two carbon atoms in ethyne.
- D The carbon-carbon triple bond in ethyne is the strongest.
- 30 Hydrocortisone is a steroid hormone produced by the adrenal gland and is released in response to stress. It is commonly used as an active ingredient in anti-inflammatory creams.



Hydrocortisone

Which of the following statements about hydrocortisone is **incorrect**?

- A It turns hot acidified K₂Cr₂O₇ from orange to green.
- B When treated with NaBH₄ in the presence of methanol, it forms a compound containing five hydroxy groups.
- C The number of chiral centres increases by 2 after it reacts with cold HCN, in the presence of NaCN.
- D When treated with an excess of hot concentrated acidified KMnO₄, it forms a compound containing five carbonyl groups.

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 31** Boron is a non-metallic element which is placed above aluminium in Group III of the Periodic Table. It forms a compound with nitrogen known as boron nitride which has a graphite structure.

Which of the following conclusions can be drawn from this information?

- 1** The empirical formula of boron nitride is BN.
- 2** The boron and nitrogen atoms are likely to be arranged alternatively in a hexagonal pattern.
- 3** Boron nitride has a layer structure with van der Waals' forces between the layers.

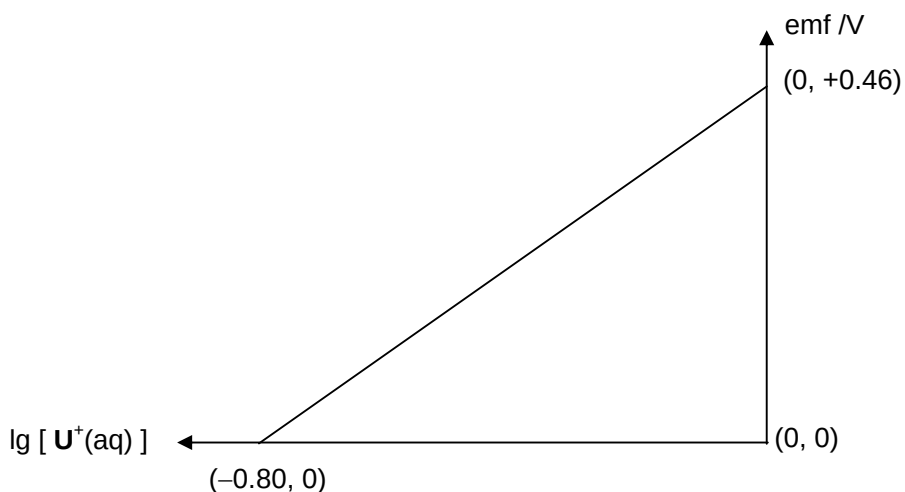
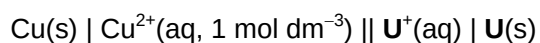
The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

32 *Use of the Data Booklet is relevant to this question.*

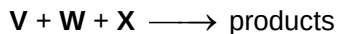
The graph below shows the variation in electromotive force (emf) of the electrochemical cell with $\lg [\text{U}^+(\text{aq})]$ at 298 K.



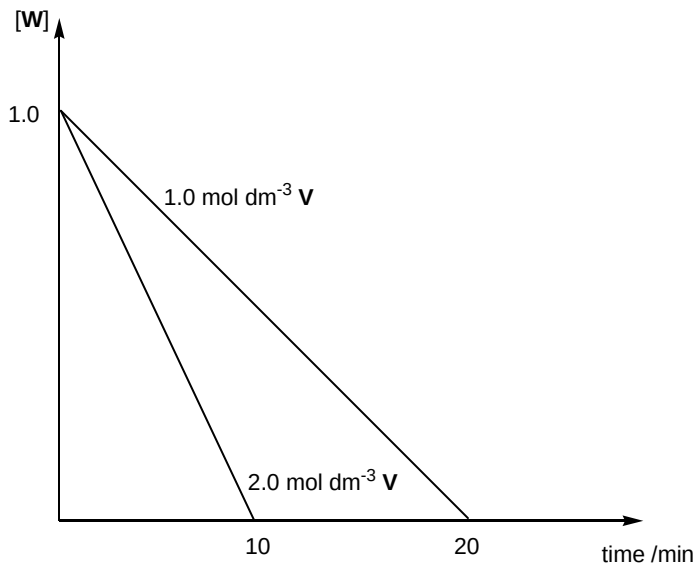
Which of the following statements are true?

- 1** Cu(s) is the negative electrode.
- 2** The emf of the given cell under standard conditions will be +0.46V.
- 3** The standard electrode potential of the $\text{U}^+(\text{aq}) \mid \text{U(s)}$ half-cell is +0.80V.

33 Consider the reaction



Two separate experiments were carried out using 1.0 mol dm^{-3} and 2.0 mol dm^{-3} of **V** respectively. A graph of the variation of the concentration of **W** with time is plotted.



From the graph, which of the following is/are **true** regarding the above reaction?

- 1 The reaction is first order with respect to **V**.
- 2 The reaction is zero order with respect to **W**.
- 3 The reaction is first order with respect to **X**.

34 Which of the following are typical properties of transition elements?

- 1 They exhibit variable oxidation states in their compounds.
- 2 All ions formed by the elements have partially filled d-orbitals.
- 3 Their atoms have fully filled d-orbitals.

35 Which of the following regarding the chlorides of the third period elements are **correct**?

- 1 All the chlorides undergo hydrolysis in water except sodium chloride.
- 2 AlCl_3 exists only in the gas phase at high temperature.
- 3 The covalent character increases.

The responses **A** to **D** should be selected on the basis of

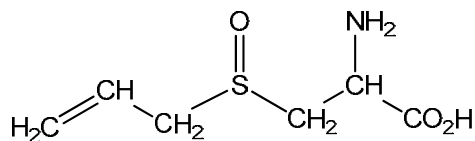
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

36 Which one of the following statements is/are likely to be true with regards to astatine or its compounds?

- 1** Astatine is a black solid at room conditions.
- 2** Hydrogen astatide decompose into its constituent elements readily.
- 3** Sodium astatide reacts with hot concentrated sulfuric acid to produce astatine.

37 Alliin is the compound responsible for the smell of garlic.



alliin

Which statements about this compound are **correct**?

- 1** It can exist in optically active forms.
 - 2** It exists as a dipolar ion in aqueous solutions.
 - 3** The bond angle O – S – C is 120°.
- 38** The reaction between carbonyl compounds and 2,4-dinitrophenylhydrazine is known as condensation. Which of the following are intermediate steps found in the multi-step mechanism of this reaction?
- 1** nucleophilic substitution
 - 2** nucleophilic addition
 - 3** elimination

- 39 Human serum albumin, HSA, is an important protein that transports long chain fatty acids and other hydrophobic molecules in the bloodstream.

Five of the most common amino acids in HSA molecule are listed below.

amino acid	formula of side chain, R in $RCH(NH_2)CO_2H$
glutamic acid	$-CH_2CH_2CO_2H$
leucine	$-CH_2CH(CH_3)_2$
lysine	$-CH_2CH_2CH_2CH_2NH_2$
valine	$-CH(CH_3)_2$
cysteine	$-CH_2SH$

Which of the following statements about the type of interactions present in HSA is correct?

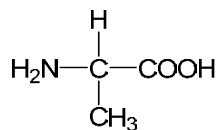
- 1 There exists hydrogen bonding between lysine and glutamic acid.
- 2 There are van der Waals' forces between leucine and valine in the protein.
- 3 There are disulfide linkages present in HSA.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

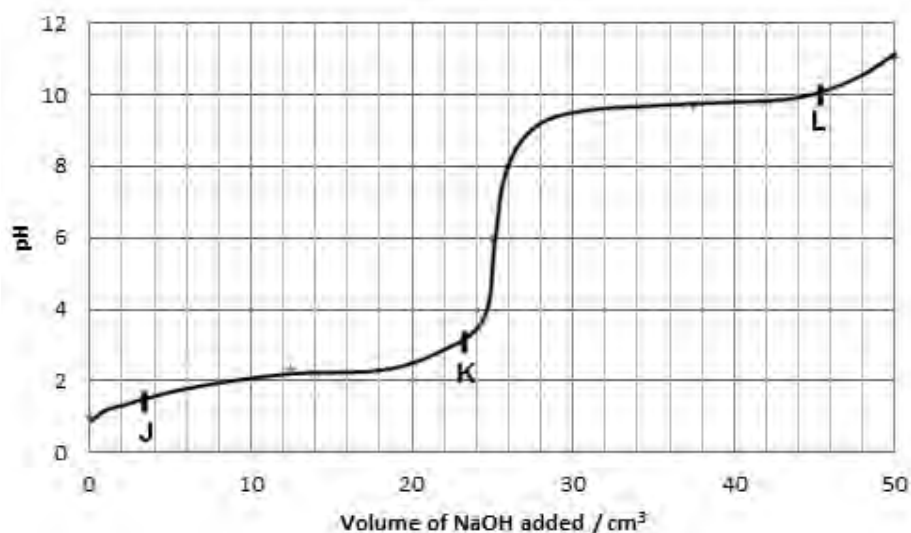
- 40** Alanine is a non-essential amino acid which is found in a wide variety of foods, especially meats.



alanine

A fully protonated alanine has 2 pKa values associated with it: 2.3 and 9.7.

The pH change when 1 mol dm⁻³ of NaOH is added to 50 cm³ of a 1 mol dm⁻³ solution of alanine in a solution of pH 1 is shown below.



Which of the following statements are true?

- 1 A buffer exists at point **L**.
- 2 The isoelectric point of alanine occurs at pH 6.

The major organic species present at point **J** and **K** are:

- 3 $\begin{array}{c} \text{H} \\ | \\ ^+\text{H}_3\text{N}-\text{C}-\text{COOH} \\ | \\ \text{CH}_3 \end{array}$ and $\begin{array}{c} \text{H} \\ | \\ ^+\text{H}_3\text{N}-\text{C}-\text{COO}^- \\ | \\ \text{CH}_3 \end{array}$ respectively